Genome-wide Aggregated Trans-Effects in Hypertension

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2026-03-06

1 Abstract

2 Introduction

Hypertension affects approximately 1.3 billion individuals worldwide and is one of the leading contributors to global mortality and disability-adjusted life-years (Guzik et al. 2024). It is a major causal risk factor for coronary heart disease, stroke, heart failure, and chronic kidney disease, and therefore contributes disproportionately to the global public health burden (Burnier and Damianaki 2023). Current pharmacological treatment relies on several drug classes that primarily target the renin-angiotensin–aldosterone system (RAAS), Ca^{2+} channels, the sympathetic nervous system, metabolic pathways and renal sodium and water handling through diuretics and related agents (Carey, Moran, and Whelton 2022), although additional physiological systems are also implicated in blood pressure regulation (Beevers, Lip, and O'Brien 2001). Despite these therapeutic options and decades of clinical experience, blood pressure control remains suboptimal: only about 18-23% of individuals with hypertension achieve guideline-recommended targets globally (NCD Risk Factor Collaboration (NCD-RisC) 2021). This persistent treatment gap highlights the need for novel, mechanistically informed targets to prevent and treat hypertension more effectively.

Human genetics offers a powerful framework for target discovery, because drug targets with robust genetic support from human studies are approximately 2.6-fold more likely to succeed in clinical development than targets lacking such evidence (Minikel et al. 2024). For blood pressure traits, large-scale genome-wide association studies (GWAS) have now identified on the order of two thousand common-variant loci (Keaton et al. 2024). However, these loci collectively explain only a small fraction of the heritability of blood pressure (around 1–3% of the phenotypic variance in individual-level studies), underscoring the highly polygenic architecture of hypertension and providing only limited direct insight into its underlying aetiology (Ehret et al. 2011). Most associated variants have small effect sizes, are non-coding, and map to broad genomic regions containing multiple candidate genes, making it challenging to pinpoint the specific genes and pathways that are truly causal and therapeutically tractable.

Recent advances in statistical genetics and the availability of large-scale genomic and molecular datasets have enabled the development of Genome-wide Aggregated *Trans* Effects (GATE) analysis, which aggregates genome-wide *trans*-acting genetic effects to prioritise "core genes" with pivotal effects in disease (Iakovliev et al. 2023). The approach has been successfully applied to oligogenic autoimmune disorders, where it has highlighted genes with clear pathogenic relevance and therapeutic potential (Iakovliev et al. 2023, 2025; Spiliopoulou et al. n.d.; Zhou et al. 2025; McKeigue et al. 2025). Extending GATE to a highly polygenic trait such as hypertension offers an opportunity to translate diffuse GWAS signals into a more interpretable set of biologically central genes and pathways. In this study, we applied GATE analysis to UK Biobank (UKB) data to identify core genes and proteins causally implicated in hypertension.

3 Methods

3.1 Data and resource availability

All analysis code is provided in our GitHub repository, and summary statistics for associations between mean arterial pressure and all GATE scores are archived on Zenodo. UK Biobank data are available on application via the UK Biobank access portal. The platform used to generate locus-specific genotypic scores, together with the database of published GWAS summary statistics, is accessible at https://genoscores.cphs.mvm.ed.ac.uk/, and access requests for the GENOSCORES platform and accompanying R package should be directed to the corresponding author.

3.2 GATE analysis

GATE analysis is a two-stage procedure, as illustrated in Figure **??**. In the first stage, we use publicly available regression coefficients (weights) for the association of genome-wide SNP data with gene expression or protein levels (quantitative trait loci or eQTL/pQTL GWAS summary statistics). In the second stage, we apply these weights to individual-level genotypes in the target dataset (UKB) to compute a locus-specific *trans* score for each region and a genome-wide aggregated *trans* (GATE) score for each transcript or protein. These scores then are tested for association with the mean arterial pressure (MAP) in the target dataset. Genes whose GATE scores are significantly associated with the phenotype ($P < 10^{-5}$), consist of effects of multiple unlinked *trans* QTLs are prioritized as putative core genes (hereafter referred to as core genes). GATE analysis methods have been reported in earlier works (Iakovliev et al. 2023, 2025; Spiliopoulou et al. n.d.; Zhou et al. 2025; McKeigue et al. 2025).

3.3 Molecular QTL Data

We used publicly available eQTL and pQTL GWAS summary statistics to derive SNP weight vectors for whole-blood transcripts and plasma proteins. These came from large-scale studies: eQTLGen Consortium Phase I (10,317 *trans* eQTLs for 6,298 genes in 31,684 participants (Võsa et al. 2021)), deCODE (4,719 proteins on the SomaLogic v4 panel in 35,559 participants (Ferkingstad et al. 2021a)), and the Olink Explore panel in 54,219 UK Biobank participants (UKB-PPP, 2,923 proteins (Sun et al. 2023)). Quality control procedures and SNP–protein coefficient estimation have been reported previously (Sun et al. 2023, 2018; Ferkingstad et al. 2021b), and 1,975 SomaLogic aptamers with evidence of cross-reactivity with CFH were excluded from the deCODE data. The eQTL and pQTL datasets were used solely to estimate SNP weights and did not contribute phenotype information.

3.4 Target Study Populations and Phenotype Definition

The UK Biobank cohort includes approximately 500,000 UK volunteers aged 40–69 years recruited between 2006 and 2010 (Bycroft et al. 2018). We used individual-level genotype and phenotype data from UKB as a target cohort, restricted to participants of European ancestry with genome-wide SNP data and valid blood pressure measurements, averaged over two assessments. Ethical approval for UKB was granted by the North West Centre for Research Ethics Committee (11/NW/0382). Written informed consent was obtained from all participants. The present study was conducted under UKB approval (application 23652).

Mean arterial pressure (MAP) was defined as two-thirds diastolic plus one-third systolic blood pressure. Hypertension is a polygenic and common condition, which affects around 30% of adults in the general population, is often diagnosed after a prolonged preclinical period, and is commonly treated with a wide range of antihypertensive and cardiometabolic medications (Irish Hypertension Society 2025; Shriner et al. n.d.; Gill et al. 2019). To reduce confounding and reverse causation, we restricted the main analysis to participants aged <60 years, in whom genetic effects on blood pressure tend to be stronger and age-related

comorbidities and treatment patterns are less likely to confound associations (Vaura et al. 2021). We also excluded individuals with common risk factors for hypertension, such as chronic kidney disease (estimated glomerular filtration rate <60 mL/min/1.73 m^2) or type 1 diabetes (Sood et al. 2024; Botdorf, Chaudhary, and Whaley-Connell 2011; Lithovius et al. 2014). When testing GATE scores derived from UKB-PPP pQTL summary statistics, individuals included in UKB-PPP were excluded from all association analyses to avoid overlap between the QTL and target datasets. We removed related individuals until no pairs had kinship coefficients > 0.05, then calculated genotypic principal components from the unrelated set. After all exclusions, we use data from 215,881 individuals, for participants receiving antihypertensive therapy, untreated blood pressure was imputed by adding 15 mmHg and 10 mmHg to systolic and diastolic blood pressure, respectively, following established CHARGE consortium recommendations (Johnson et al. 2011).

3.5 Computation of GATE Scores in the Target Dataset

For each genomic region with at least one SNP associated with a transcript or protein at $P < 10^{-6}$, we defined a quantitative trait locus by grouping all SNPs associated with that transcript or protein at $P < 10{-}5$. A locus was classified as *cis* if its distance from the transcription start site of the encoding gene was < 50 kb, *cis-x* if 50 kb to 5 Mb, and *trans* if > 5 Mb. To obtain linkage disequilibrium (LD)-adjusted SNP weights, we premultiplied the univariate regression coefficients by the inverse genotype covariance matrix (add 1000 genomes reference). For each individual, the GATE score for a transcript (eGATE score) or protein (pGATE score) was then calculated as the dot product of the genotypes in the target cohort and these LD-adjusted weights at *trans* QTLs, representing the genetically determined expression or protein level from all *trans* QTLs. The locus-specific contributions are reported in Supplementary Tables **??**, **??**. Similarly to previous studies, to avoid confounding by large direct effects on disease, we excluded *trans* eQTLs in the HLA region from GATE scores. To account for the polygenicity of hypertension, for pQTLs we additionally excluded loci such as UMOD, APOE, ABO, and SH2B3, which act as pervasive master regulators of many plasma proteins.

3.6 Association Analysis in UKB and Selection of Core Genes

We tested associations between GATE scores and MAP in unrelated UKB participants of European ancestry, consistent with the European-ancestry restriction of the eQTL and pQTL studies used in the first step. For each score, we fitted a linear regression model with MAP as the continuous outcome and the GATE score as the predictor, adjusted for age, sex, body mass index, body fat percentage, self-reported ethnic background, alcohol consumption, autoimmune thyroid deficiency and 40 genotype principal components. The covariates were chosen based on a regression model seeking association to determine what were the main covariates in the UKB dataset that are associated with blood pressure variation. We adjusted for measures of adiposity because we were explicitly interested in genetic determinants of hypertension beyond those mediated by obesity. We also excluded individuals with autoimmune thyroid disease to reduce potential confounding by autoimmunity (Berta et al. 2019). *Cis* score associations with MAP were tested separately.

Putative core transcripts or proteins (collectively termed core genes for simplicity) were selected using two criteria: GATE score association with MAP at $P < 10^{-5}$ and effective number of *trans* loci. To reduce the influence of a single pleiotropic variant and maximize signal-to-noise, we required that each GATE score be supported by multiple independent *trans* loci. We quantified this using a diversity index that measures the effective number of loci contributing to the *trans* score and applied thresholds of ≥ 5 (standard GATE) and ≥ 8 or eQTLs or ≥ 20 for pQTLs for the stringent sets in the main text. The formal definition and justification of this index are given in Supplementary Methods.

3.7 Additional Evidence Supporting Causality for Associated GATE Scores

To further support a causal role for core genes, firstly, we analysed the UKB dataset for *cis* variants (within 5 Mb of the transcription site) associated with MAP or GWAS hits for essential hypertension within 200 kb of

the transcription start site are reported in the published meta-GWAS (Cerezo et al. 2025). And secondly, we conducted a Mendelian randomization to test for dose-response relationship between the effect of the *trans* QTLs on the expression of the gene or protein and the effect of those same *trans* QTLs on blood pressure. See Supplementary Material for details.

On the other hand, we further sought external data for additional evidence to support a causal role for core genes. We examined independent evidence including: (1) blood pressure-related or cardiovascular/metabolic disease caused by rare variants in the gene or pathway; (2) blood pressure associations ($n = 38,043$) with measured protein levels in UKB-PPP, expected to be stronger than GATE associations if causal; (3) published functional evidence that gene knockout, overexpression, or pharmacological perturbation alters blood pressure in experimental models; and (5) drugs targeting the gene product, ligand, or receptor that affect blood pressure.

4 Results

4.1 Association of mean arterial pressure with GATE scores for gene expression

Table **??** presents the 28 GATE scores associated with mean arterial pressure at $P < 10^{-5}$, where the effective number of *trans* eQTLs was at least 8.

As shown in the table, there were 28 genes that met these filtering criteria, with TREM1 being the most strongly associated. For the majority of these genes there was no evidence of an association of *cis* scores in that gene with MAP, with borderline significant associations for *cis* scores in *BANK1*, *DPYSL2* and *GPBAR1*. Of the 28 putative core genes, coregenes.eqtl.strict[!is.na(reported.genes),.N] have previous GWAS hit for essential hypertension with 200kb, most of which were not localised to any specific gene in the originating GWAS.

Figure **??**, displays a heatmap of correlations between GATE scores based on aggregated *trans* eQTLs ordered by hierarchical clustering. Strong correlations between scores indicate substantial sharing of *trans* eQTLs contributing to the scores. (Table **??**). As shown there is a block of highly correlated scores within which there is a subcluster. Annotation of the known functions of these genes shows these genes are mostly in immune pathways with the subcluster being B-cell related genes. To show sharing of *trans* eQTLs , we grouped overlapping *trans* eQTLs into clumps and listed all genes they contribute to as tabulated in Supplementary Table **??** . As highlighted in bold, many clumps harbor known GWAS hits for hypertension and contribute to many core genes (i.e. are " peripheral master regulators").

4.2 Association of mean arterial pressure with GATE scores for circulating proteins

The GATE analysis based on circulating protein levels identified 24 pGATE scores associated with mean arterial pressure at $P < 10^{-5}$, following filtering by the effective number of *trans* pQTLs at least 20. A more stringent criterion for effective number of *trans* pQTLs was used to limit the data presentation to a reasonable size. In Table **??**), the most strongly associated pGATE scores was for *IL17RB*. For none of these 24 genes was there any evidence of an association of *cis* scores in that gene with MAP. Of the 24 putative core genes shown, 11 had a prior GWAS hit within 200kb most of which could not be localised to any gene in the originating GWAS.

As illustrated in Figure **??**, the GATE scores were relatively uncorrelated, suggesting that these genes are not linked by a shared biological pathway. This is consistent with the minimal sharing of *trans* pQTLs observed between these genes (Supplementary Table **??**).

A full listing of all eGATE and pGATE scores N=210 that were associated with MAP at $P < 10^{-5}$ but where the effective number of loci threshold is relaxed to ≥ 5 is shown in Supplementary Tables **??** and **??**. Of the total of 210 genes where eGATE or pGATE scores were associated with MAP at this threshold 94 harbored a prior GWAS hit within 200kb most of which could not be localised in the originating GWAS.

4.3 Instrumental variable analysis for supportive evidence of causality (Mendelian randomization analysis)

Supplementary Table **??** details the results of the MR analysis for putative core genes listed in Table **??** and Table **??**. We considered that an effective number of loci needs to be at least 20 for MR analysis to have power to demonstrate supportive evidence of causality for any putative core gene. None of the eGATE derived putative core genes had this number of effective loci, and of these only for ABCA1 was there a significant MR result at $P < 0.01$. For the 24 putative core genes based on pGATE scores in Table **??**, by the filtering criterion all had an effective number of loci at least 20. Altogether 13 of 24 genes in Table **??** had MR support for causality at $p < 0.01$ (i.e. marginalizing over pleiotropic effects supported a dose-response effect of the protein on MAP at $P < 0.01$).

Figure **??** displays the *trans* pQTL-disease associations plotted against the *trans* pQTL-transcript associations for the top 6 genes with MR support for causality: *IL17RB*, *CECAM16*, *PTPRF*, *CKB*, *TCL1A* and *CNTN3*.

4.4 Other Supportive Evidence for the Role of Putative Core Proteins for MAP

In addition, Table **??** summarizes for each putative core gene the results of the prespecified criteria other than *trans* QTL effects that support causality of the gene for MAP. In addition to 24 genes having known GWAS hits or *cis* associations, 13 having significant MR support we note that 1. Core genes have previously been reported to harbor variants for monogenic forms of disease that could affect blood pressure: *ABCA1*, *ADIPOQ*, *DSG2*, *ELANE* and *EDA2R*. 2. For 17 of the core genes, measured levels of the proteins encoded by these genes in the UKB-PPP study showed associations with MAP at $P < 10^{-4}$ in Table **??**. The directions of these associations with MAP were consistent with the GATE analysis for 11 core genes. See Supplementary Table **??** for the detailed results. 3. For 9 of the core genes, we identified supportive evidence for a causal relationship based on perturbation in a mouse model of hypertension. Details are summarized in Supplementary Table **??**. For xx genes (TODO:LIST them) the directions of effect of the genetically predicted levels of gene expression (for eGATE scores) or protein levels (for pGATE scores) on MAP in the GATE analysis are the same as that expected from animal models. For xx genes, TODO:LIST THESE their directions of effect in the GATE analysis are the opposite to that expected from animal models. 4. For one gene the proteins encoded by these genes or other proteins in the known pathway these proteins are involved in are in drug development for conditions that affect blood pressure: *TREM1*.

5 Discussion

Applying GATE to mean blood pressure without imposing strong prior hypotheses implicates, alongside established mechanisms involving endothelial function, neural regulation, kidney function and the RAAS, a broad array of immune-related genes. These findings are concordant with accumulating experimental and clinical evidence over recent decades that positions immune dysregulation, encompassing both innate and adaptive immune responses as an important contributor to the development and maintenance of elevated blood pressure (Guzik et al. 2024). Nevertheless, immune pathways remain relatively underexplored as therapeutic targets in hypertension compared with their prominence in other cardiovascular and autoimmune conditions, suggesting that a genetics-led approach may help to uncover tractable immune and non-immune mechanisms for blood pressure control (Murray et al. 2021).

5.1 Immune mechanisms in hypertension

Hypertension is increasingly recognised as a state of chronic, low-grade inflammation driven by coordinated activation of both the well-established innate immune and the less well-characterised adaptive immune responses (Guzik et al. 2024). The causal contribution of adaptive immunity is supported by animal

studies showing that mice lacking mature T and B cells, or specifically $\gamma\delta$ T cells, are protected from blood pressure elevation across several experimental models (Guzik et al. 2007; Caillon et al. 2017). In humans, higher circulating levels of $\gamma\delta$ T cells and NK cells associate with higher systolic blood pressure, while pro-inflammatory cytokines such as IL6, TNF-α, IL1β, IL18 and CCL2 prospectively predict incident hypertension (Guzik et al. 2024; Delaney et al. 2021). Together with evidence of increased IgG production and vascular IgG deposition in angiotensin II-induced hypertension, and the presence of pro-inflammatory and immunosenescent T cell subsets even in children with primary hypertension, these findings support a model in which T and B cells, autoantibodies, NK cells and myeloid antigen-presenting cells orchestrate vascular and renal inflammation, promoting increased vascular resistance and target-organ damage (Harrison and Patrick 2024; Gackowska et al. 2018).

Genetic and transcriptomic data converge on a similar picture of immune-driven hypertension. Integrative network analysis in the Framingham Heart Study identified SH2B3 as a major driver of blood pressure, with link to T cell activation networks and extensive transcriptional reprogramming in Sh2b3-deficient mice (Huan, Meng, et al. 2015; Saleh et al. 2015). Gene expression meta-analysis has implicated cytotoxic lymphocyte genes (FGFBP2, PRF1, GNLY, NKG7) and T cell activation genes (IL2, IL2RA, IL2RB, IL15) in blood pressure regulation (Huan, Esko, et al. 2015). Complementary GWAS signals at co-stimulatory molecules CD80, CD86 and CD247 further point to co-stimulation and T cell receptor signalling as key pathways (Pineda et al. 2020). Renal multi-omics studies add end-organ context, revealing dysregulation of interferon signalling, TNF-mediated pathways and proteasomal antigen processing in hypertensive kidneys (Eales et al. 2021).

The present GATE analysis prioritises a set of core genes that are enriched for immune functions, including cytokines, inflammasome components and co-stimulatory molecules involved in T cell activation, cytotoxic effector function and antigen presentation. By converging on immune-related core genes from a purely genetic perspective, GATE provides orthogonal support for the central role of immune dysregulation in hypertension and offers a genetically anchored framework for prioritising immune pathways as potential therapeutic targets.

5.2 Genes with convergent evidence of causality

TODO

5.3 Biological interpretation

5.3.1 Genes that act on or are expressed by endothelial cells

- *CD300LG* is expressed predominantly on vascular endothelial cells. It has recently been shown to act as a receptor for triglyceride-rich lipoproteins (chylomicrons and VLDL) that facilitates postprandial lipid clearance (Granade et al. 2025). Both the GATE score for CD300LG and measured levels of the protein are inversely associated with blood pressure, and also with plasma triglyceride, coronary disease and type 2 diabetes [show forest plot in supplementary material].
  - Physical activity increases CDL300LG levels (Lee-Ødegård et al. 2024).
- *CD36* encodes a scavenger receptor that mediates transport of nonesterified fatty acids across the endothelial cell barrier (Son et al. 2018) through packaging with ceramides into exosome-like small extracellular vesicles that are released at the basolateral membrane (Peche et al. 2023).

Though the GATE score for *CD36* is positively associated with blood pressure, CD36 deficiency in humans is associated with hypertension and with metabolic disturbances including elevated triglyceride levels and resistance to insulin-mediated glucose uptake (Hirano et al. 2003)

- *ADIPOQ* encodes adiponectin, produced by adipocytes and circulates at micromolar concentrations. Adiponectin enters endothelial cells via CDH13 receptor and forms exosomes containing ceramides - it appears to act as a garbage collector (Tanaka et al. 2019; Obata et al. 2018).

A possible mechanism by which altered lipid handling in vascular endothelial cells might raise blood pressure is through accumulation of lipid droplets (Jaffe and Karumanchi 2024). Endothelial lipid droplet accumulation in response to high-fat diet reduces nitric oxide production and raises blood pressure (Kim et al. 2023).

- DSG2, expressed on endothelial cells, encodes a receptor for the fatty acid binding protein FABP4 (Chen et al. 2023). FAB4 levels are associated with blood pressure and the GATE score for FABP4 is associated with blood pressure (probably not in table 2 because below threshold).

5.3.2 Genes in immune / inflammatory pathways

- Inverse association of *IL17RB* with blood pressure (also with coronary disease). *IL17RB* is broadly expressed, it encodes the receptor for IL-17B, and forms a heterodimer with IL-17RA that is the receptor for IL-25 (alias IL-17E). IL-25 signalling via this pathway induces the expression of *CXCL8* (alias IL-8). No experimental studies of effects of IL-17RB perturbation on blood pressure have been reported. An IL-17RB inhibitor has reached clinical stage for atopic dermatitis (Xu et al. 2024).
- *BTN2A1* is an immune checkpoint targeting V 9V 2 T cell cytotoxicity
- *CLEC4C* receptor for asialo-galactosyl-oligosaccharides. It inhibits interferon production by plasmacytoid dendritic cells
- *CD209* also known as dendritic cell-specific intercellular adhesion molecule-3–grabbing nonintegrin (DC-SIGN), is a C-type lectin receptor that is expressed primarily on subsets of dendritic cells and certain macrophages. It recognizes and binds to specific carbohydrate moieties—such as high-mannose structures and fucose-containing glycans—on a broad range of pathogens.
- *CKB* creatine kinase B plays an unexpected role in modulating TCR-mediated signaling and critically regulates thymocyte selection and T cell activation.
- *BCL11A* selective interactions with the NR2E/F subfamily of orphan nuclear receptors, essential for the normal development of B and T cells
- *BANK1* promotes tyrosine phosphorylation of inositol 1,4,5-trisphosphate receptors, expressed in B cells
- *MEF2C* expressed by NK cells, encodes a transcription factor that regulates NK cell function via effects on lipid metabolism
- *TREM1* encodes a receptor belonging to the Ig superfamily that is expressed on myeloid cells. This protein amplifies neutrophil and monocyte-mediated inflammatory responses
- *ALOX5AP* encodes a protein which, with 5-lipoxygenase, is required for leukotriene synthesis
- COBLL1 expression changes during B-cell maturation in non-malignant secondary lymphoid tissue with a higher expression in germinal center B cells than naïve and memory B cells.
- TNFRSF13B encodes TACI, a B cell activating receptor
- IL2RB is one component of the alpha/beta/gamma subunit heterotrimer that is the high-affinity receptor for IL-2. Low-dose IL-2 normalized multi-organ mt function and hypertension in response to placental ischemia.

5.3.3 Genes with effects on RAAS or kidney

- *CPA1* encodes carboxypeptidase 1 which cleaves dietary proteins but also degrades angiotensin II (Pereira et al. 2012).
- RTN1 encodes reticulon-1, and overexpression in renal cells induces ER stress and apoptosis. Knockdown of Rtn1a expression *in vivo* attenuates ER stress, proteinuria, glomerular hypertrophy and mesangial expansion in diabetic mice.
- *KCNJ15* encodes the inwardly rectifying potassium channel Kir4.2, expressed in multiple tissue, specifically in the collecting ducts of the kidney.

5.3.4 Genes with effect on lipids / adiposity

- COBLL1 – ligand for ROR1 receptor
  Common variant affects metabolically obese normal weight phenotype by reducing ability of subcutaneous adipocytes to store fat. also expressed in Renal epithelial cells - Ion transport and pH regulation (mainly)
- ABCA1

5.4 Limitations

- Limitations of the analysis platform
  - we have summary stats with > 5 trans-QTLs for only about 500 transcripts and 3500 plasma proteins. This should improve within next year or so.
  - No coverage of transcripts in tissues other than whole blood. Plasma proteins appear to cover many proteins that have tissue-specific expression
- cannot control directly for cell type proportions in whole blood
- Limitations of the method:
  - we can't easily exclude reverse causation, as we can't establish temporal sequence. This would require summary stats from younger age groups before hypertension becomes common. We can do a sensitivity analysis to exclude effects of medication.
  - Pleiotropy from direct *trans* QTL effects on disease may confound the inference of core genes, where a gene may appear associated with disease solely because its GATE score contains *trans* QTLs from regions with large direct effects.
- GATE scores for gene expression explain only approximately 2% of transcript variance, whereas the estimated *trans* heritability is around 40%. The true effects are substantially larger than those detected. Nevertheless, even weak genetic proxies can tag genes with large effects on disease, as shown previously (Iakovliev et al. 2023, 2025; Spiliopoulou et al. n.d.; Zhou et al. 2025; McKeigue et al. 2025).

6 Conclusions

7 Acknowledgements

8 Funding

SB and AI were supported by MRC Cross-Disciplinary fellowships: add more

9 Data availability

add usual statement about ukbb

10 Author contributions

fill this out

11 Guarantor Data Access and Responsibility Statement

12 Duality of Interest

No potential conflicts of interest relevant to this article were reported.

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Figures

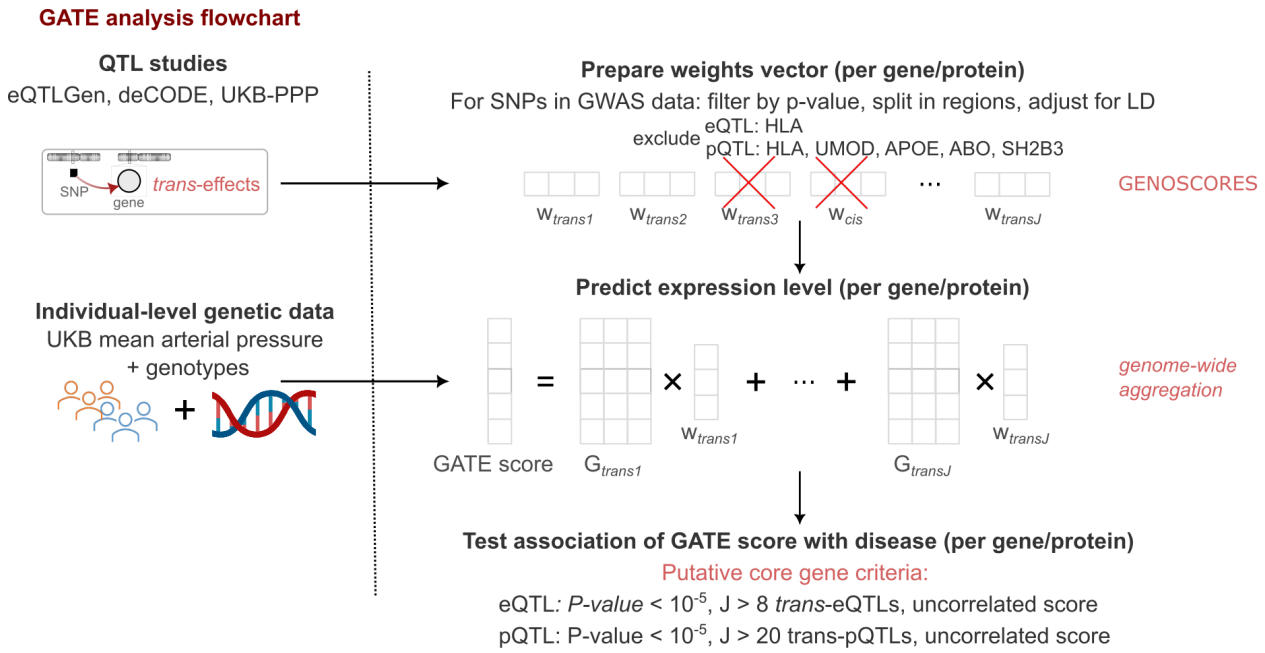

Figure 1: Schematic overview of the GATE analysis. Left (inputs): summary-level eQTL and pQTL GWAS results together with individual-level genotypes and case/control status from the target cohorts. Right (analysis workflow): GATE scores are derived from the eQTL and pQTL summary statistics and individual-level genotypes and then tested for association with mean arterial pressure. For transcripts and proteins whose scores indicate core genes, additional lines of evidence are assessed to support a causal role of the corresponding gene in disease.

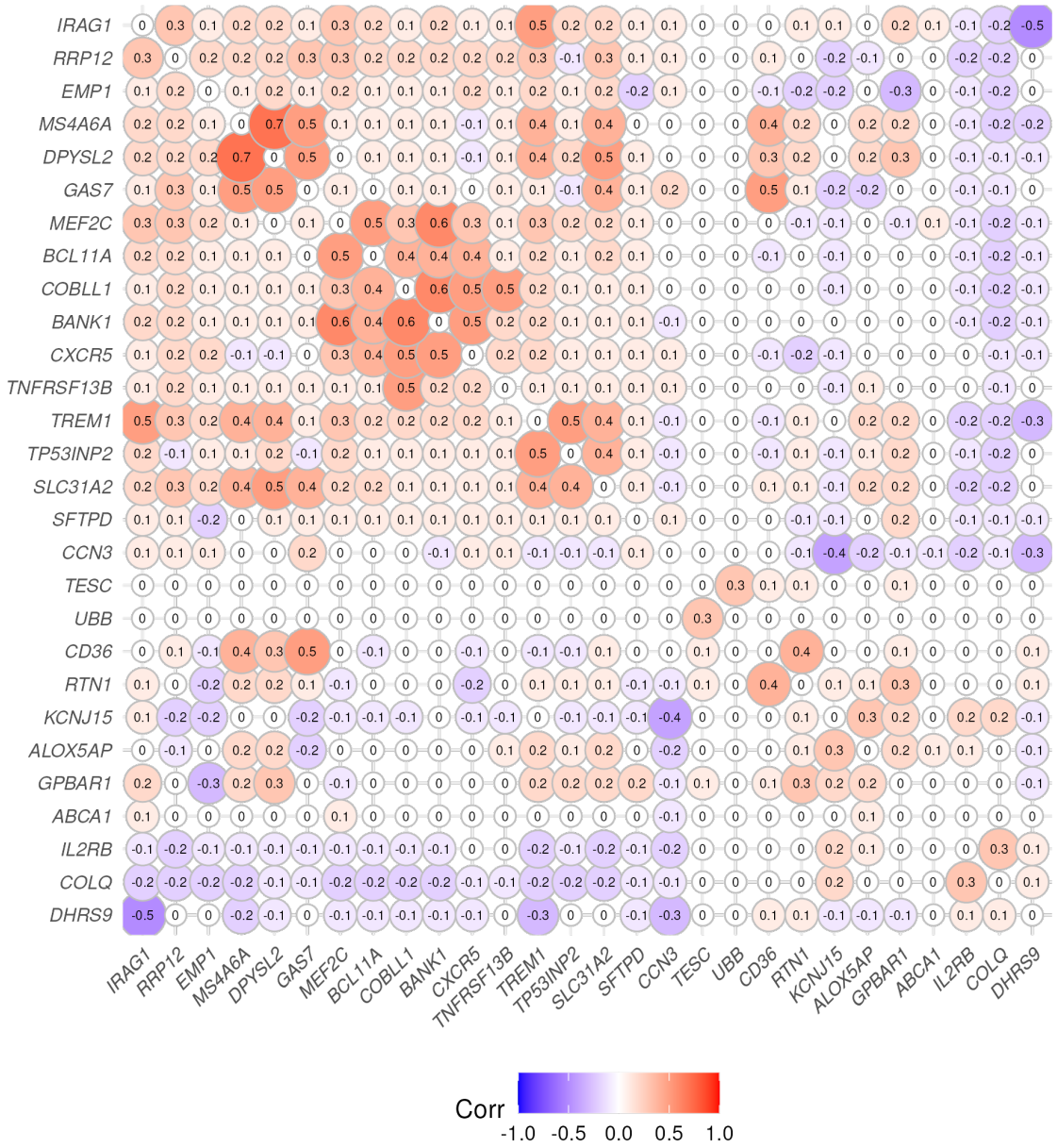

Figure 2: Heatmap of pairwise correlations among eGATE scores for putative core transcripts in MAP, arranged by hierarchical clustering. Our reporting criteria for putative core genes were strength of association ($P < 10^{-5}$) with at least eight effective *trans* eQTLs contributing to the eGATE score. Two distinct clusters emerge: a block of highly correlated scores is evident, containing a subcluster of B-cell-related genes; the broader block comprises genes predominantly involved in immune pathways.

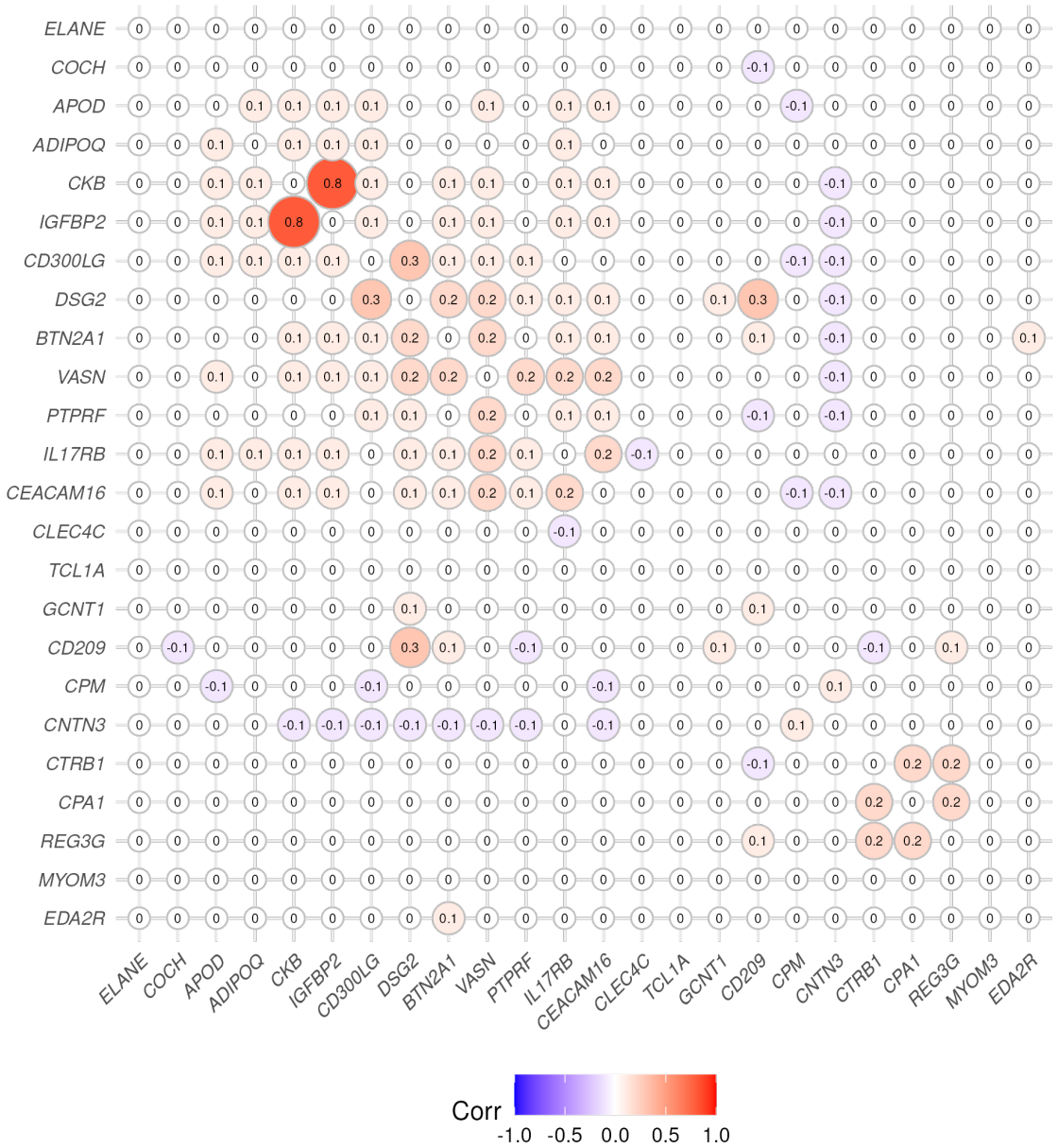

Figure 3: Heatmap of pairwise correlations among pGATE scores for putative core proteins in MAP, arranged by hierarchical clustering. Our reporting criteria for putative core genes were strength of association ($P < 10^{-5}$) with at least twenty effective *trans* pQTLs contributing to the pGATE score. The pGATE scores were largely independent of one another, indicating that the corresponding genes do not converge on a common biological pathway.

Tables

Table 1: Chr, chromosome; Mb, megabase; β, regression coefficient representing the change in MAP (mmHg) per SD increase in the eGATE score. For the analysis 212,569 individuals were included. Putative core genes were defined by an association $P < 10^{-5}$ and minimum of eight effective *trans* eQTLs contributing to the eGATE score. Reported GWAS hits for essential hypertension are those falling within 200 kb of the transcription start site; NR indicates that the SNP has not been attributed to a specific gene.

	Transcription site		*trans* score			*cis* score		
Gene	Chr	Start position (Mb)	Effective number of *trans* eQTLs	β	P value	β	P value	Reported GWAS hit
TREM1	6	41.27	8.1	-0.019	5×10^{-25}	-0.003	0.2	**NR**
SFTPD	10	79.94	9.2	-0.015	6×10^{-16}	0.001	0.7	**NR**
SLC31A2	9	113.15	9.4	-0.013	6×10^{-13}	0.000	0.9	**.**
DHRS9	2	169.06	8.6	0.012	3×10^{-11}	0.002	0.3	**NR**
BANK1	4	101.41	9.9	-0.012	3×10^{-11}	0.007	3×10^{-4}	**BANK1**
IL2RB	22	37.12	9.8	0.012	1×10^{-10}	-0.003	0.08	**NR**
TP53INP2	20	34.70	8.1	-0.012	6×10^{-10}	0.002	0.2	**.**
IRAG1	11	10.57	8.5	-0.011	8×10^{-10}	-0.002	0.4	**NR**
TNFRSF13B	17	16.93	14.6	-0.011	1×10^{-8}	0.002	0.3	**.**
KCNJ15	21	38.16	8.2	0.011	1×10^{-8}	-0.002	0.2	**.**
EMP1	12	13.20	10.2	-0.011	2×10^{-8}	0.001	0.5	**NR**
COLQ	3	15.45	8.9	0.010	2×10^{-8}	0.002	0.4	**NR**
DPYSL2	8	26.51	8.8	-0.010	1×10^{-7}	-0.009	3×10^{-6}	**.**
RTN1	14	59.60	9.3	0.010	2×10^{-7}	-0.002	0.2	**.**
COBLL1	2	164.65	9.7	-0.010	2×10^{-7}	-0.005	0.02	**.**
CCN3	8	119.42	13.8	-0.010	2×10^{-7}	-0.002	0.4	**.**
RRP12	10	97.36	8.3	-0.010	3×10^{-7}	0.000	0.8	**.**
MS4A6A	11	60.17	9.5	-0.009	4×10^{-7}	0.002	0.4	**MS4A6A**
ALOX5AP	13	30.71	8.1	-0.009	8×10^{-7}	0.000	0.8	**.**
UBB	17	16.38	15.1	0.009	1×10^{-6}	.	.	**.**
BCL11A	2	60.45	9.9	-0.009	2×10^{-6}	0.001	0.7	**.**
ABCA1	9	104.78	10.0	-0.009	2×10^{-6}	0.001	0.7	**.**
CXCR5	11	118.88	10.5	-0.009	2×10^{-6}	-0.003	0.1	**.**
GPBAR1	2	218.26	10.4	-0.009	3×10^{-6}	0.007	3×10^{-4}	**.**
TESC	12	117.04	9.9	0.009	4×10^{-6}	0.002	0.3	**.**
MEF2C	5	88.72	8.5	-0.009	4×10^{-6}	0.001	0.8	**NR, LINC00461**
CD36	7	80.37	9.3	0.008	5×10^{-6}	0.004	0.02	**NR**
GAS7	17	9.91	13.9	-0.008	6×10^{-6}	0.001	0.6	**.**

Table 2: DeC, deCODE study; UKB, UK Biobank Pharma Proteomics Project; Chr, chromosome; Mb, megabase; β, regression coefficient representing the change in MAP (mmHg) per SD increase in the pGATE score. For the analysis 212,569 individuals were included from DeC and 190,635 individuals from UKB, with participants enrolled in the proteomics sub-study excluded. Putative core genes were defined by an association $P < 10^{-5}$ and minimum of twenty effective *trans* pQTLs contributing to the pGATE score. Reported GWAS hits for essential hypertension are those falling within 200 kb of the transcription start site; NR indicates that the SNP has not been attributed to a specific gene. Where multiple pGATE scores exist for a given protein (e.g., derived from different pQTL GWAS), only the score with the strongest association is reported.

		Transcription site		*trans* score			*cis* score		
pQTL study	Gene	Chr	Start position (Mb)	Effective number of *trans* pQTLs	β	P value	β	P value	Reported GWAS hit
UKB	*IL17RB*	3	53.85	23.6	-0.016	3×10^{-17}	-0.004	0.03	**NR, CACNA1D**
UKB	*ADIPOQ*	3	186.84	20.7	-0.015	3×10^{-13}	0.000	1	**NR**
UKB	*PTPRF*	1	43.53	25.6	-0.014	3×10^{-12}	0.004	0.06	**NR**
UKB	*CD300LG*	17	43.85	23.6	-0.014	4×10^{-12}	-0.001	0.8	**.**
UKB	*CKB*	14	103.52	20.7	-0.014	1×10^{-11}	.	.	**NR**
UKB	*CEACAM16*	19	44.70	24.0	-0.013	2×10^{-11}	.	.	**NR, APOE, PVRL2, TOMM40, APOC1, BCAM**
UKB	*CNTN3*	3	74.26	33.5	0.013	3×10^{-11}	-0.001	0.5	**.**
UKB	*IGFBP2*	2	216.63	21.7	-0.013	3×10^{-11}	-0.001	0.5	**.**
UKB	*APOD*	3	195.57	20.5	-0.012	8×10^{-10}	-0.003	0.2	**.**
UKB	*DSG2*	18	31.50	20.2	-0.012	1×10^{-9}	0.000	0.9	**.**
UKB	*VASN*	16	4.37	27.0	-0.012	2×10^{-9}	0.001	0.5	**.**
UKB	*GCNT1*	9	76.42	22.3	-0.011	2×10^{-8}	0.002	0.2	**.**
DeC	*CTRB1*	16	75.22	20.2	-0.010	4×10^{-8}	0.005	0.03	**NR, LOC105371345**
UKB	*CPM*	12	68.84	21.5	0.011	5×10^{-8}	0.000	1	**.**
UKB	*CLEC4C*	12	7.73	24.8	0.010	2×10^{-7}	0.001	0.5	**NR**
UKB	*TCL1A*	14	95.71	35.7	0.010	3×10^{-7}	0.004	0.07	**.**
UKB	*CPA1*	7	130.38	29.2	-0.010	4×10^{-7}	0.000	0.8	**NR**
UKB	*MYOM3*	1	24.06	24.6	-0.010	8×10^{-7}	0.001	0.6	**.**
DeC	*ELANE*	19	0.85	21.3	0.009	8×10^{-7}	0.001	0.6	**.**
UKB	*EDA2R*	X	66.60	21.2	-0.010	9×10^{-7}	.	.	**.**
UKB	*BTN2A1*	6	26.46	22.2	0.009	3×10^{-6}	.	.	**NR, HFE**
UKB	*CD209*	19	7.74	25.9	0.009	6×10^{-6}	0.001	0.8	**NR**
DeC	*COCH*	14	30.87	20.7	-0.008	9×10^{-6}	0.002	0.3	**.**
DeC	*REG3G*	2	79.03	24.7	-0.008	1×10^{-5}	0.001	0.6	**NR**

Table 3: Additional evidence in support of GATE-detected core genes for MAP with thresholds $P < 10^5$, effective numbers of eQTL ≥ 8 and pQTLs ≥ 20.
'+' indicates that the criterion is met; '—' indicates that it is not; '.' data is missing.
(a) Known GWAS hit for essential hypertension exists within 200 kb or if a textitcis-score for the gene is associated with MAP ($P < 0.001$).
(b) Mendelian randomization based on textittrans-QTLs and marginalizing over the distribution of pleiotropic effects supports a causal effect for the gene on MAP ($P < 0.01$ and effective number of eQTLs/pQTLs ≥ 20).
(c) Rare variants in the gene cause monogenic form of the blood pressure-related disease.
(d) Correspondence between measured protein levels in UKB-PPP and MAP: '+' indicates association at $P < 10^{-4}$ with effect direction consistent with GATE score effect on MAP; '-' indicates opposite direction; '0' indicates no association.
(e) There is evidence of causality through perturbation of the gene in an animal model of hypertension or related phenotype.
(f) Existing investigational drug that targets the protein encoded by the gene or its ligand/receptor has some clinical evidence of efficacy against blood pressure-realated disease.

Gene symbol	GWAS hit[a]	Mendelian randomization[b]	Monogenic cause[c]	Measured gene product[d]	Experimental validation[e]	Drug effect[f]
TREM1	+	.	-	.	-	+
SFTPD	+	.	-	0	-	-
SLC31A2	-	.	-	.	-	-
DHRS9	+	.	-	.	-	-
BANK1	+	.	-	-	-	-
IL2RB	+	.	-	0	-	-
TP53INP2	-	.	-	.	+	-
IRAG1	+	.	-	.	+	-
TNFRSF13B	-	.	-	0	-	-
KCNJ15	-	.	-	.	-	-
EMP1	+	.	-	.	-	-
COLQ	+	.	-	.	-	-
DPYSL2	+	.	-	.	-	-
RTN1	-	.	-	.	+	-
COBLL1	-	.	-	.	-	-
CCN3	-	.	-	-	-	-
RRP12	-	.	-	.	-	-
MS4A6A	+	.	-	.	-	-
ALOX5AP	-	.	-	.	-	-
UBB	-	.	-	.	-	-
BCL11A	-	.	-	.	-	-
ABCA1	-	.	+	.	-	-
CXCR5	-	.	-	.	-	-
GPBAR1	+	.	-	.	-	-
TESC	-	.	-	.	-	-
MEF2C	+	.	-	.	-	-
CD36	+	.	-	+	-	-
GAS7	-	.	-	.	-	-
Analysis based on circulating proteins						
IL17RB	+	+	-	+	+	-
ADIPOQ	+	-	+	0	+	-
PTPRF	+	+	-	0	-	-
CD300LG	-	-	-	+	+	-
CKB	+	+	-	+	+	-
CEACAM16	+	+	-	0	-	-
CNTN3	-	+	-	+	-	-
IGFBP2	-	+	-	+	+	-
APOD	-	+	-	0	-	-
DSG2	-	-	+	+	-	-
VASN	-	-	-	0	-	-
GCNT1	-	+	-	0	-	-
CTRB1	+	-	-	-	-	-
CPM	-	+	-	+	-	-
CLEC4C	+	-	-	+	-	-
TCL1A	-	+	-	+	-	-
CPA1	+	-	-	-	-	-
MYOM3	-	-	-	-	-	-
ELANE	-	-	+	.	+	-
EDA2R	-	+	+	+	-	-
BTN2A1	+	+	-	0	-	-
CD209	+	-	-	0	-	-
COCH	-	+	-	-	-	-
REG3G	+	-	-	0	-	-